Aziz Mohammad

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EDUCATION

University of Illinois Chicago, Chicago, IL — Aug 2025 – May 2027 (Expected)
BS in Mechanical Engineering
GPA – 3.8/4.0 | Technical GPA – 3.86/4.0 | Dean's List
Relevant Coursework: Heat Transfer, Fluid Mechanics, Thermodynamics 1&2
College of DuPage, Glen Ellyn, IL — Aug 2023 – May 2025

WORK EXPERIENCE

Wilo USA, *Systems Engineering Intern***,** Rosemont, IL — May 2026 – August 2026

- Led mechanical design of a modular family of commercial HVAC hydronic pump packages (30–230 tons), converting custom one-off skids into 20+ standardized product configurations to establish a repeatable product line.
- Integrated mechanical, hydraulic, and electrical systems into a unified skid-mounted package, designing 5" grooved piping loops, thermal insulation, electrical conduit routing, and dual-pump configurations with VFDs, expansion vessels, buffer tanks, air/dirt separators, and single-point power entry.
- Built structured configuration matrices, automated Excel costing models, and standardized hardware datasets across international DIN/US standards, eliminating redundant drafting tasks and significantly accelerating sales quotation response times.

University of Illinois Chicago, *Research Assistant at Intelligent Multiphase Systems Lab* — March 2026 – Present

- Developing a precision flow-boiling experimental apparatus to investigate multiphase heat transfer and interfacial transport physics; utilizing SolidWorks for complex CAD modeling and advanced fabrication to ensure high-fidelity data acquisition for electronics cooling.
- Designed and fabricated custom heater insulation for vapor chamber testing using SLA 3D printing; optimized resin-based prints to meet precise thermal and structural requirements for laboratory experimentation.

CLUBS AND ACTIVITIES

Society of Automotive Engineers (SAE), *Member* — Aug 2025 – Present

- Optimized a custom FSAE push bar using SolidWorks and Ansys FEA, achieving a **20%** weight reduction while maintaining a **2.5x** factor of safety to enhance vehicle agility and meet stringent competition structural requirements.
- Engineered and fabricated a precision welding jig for the vehicle's tubular space frame, maintaining **±1.5 mm** tolerances to ensure perfect chassis symmetry and the alignment of critical suspension hard points.

American Society of Mechanical Engineers (ASME), *Member* — Aug 2025 – Present

- Collaborated on cross-disciplinary engineering teams to develop hardware solutions using mechanical principles and CAD modeling, contributing to technical design reviews and peer-reviewed projects focused on industry-standard manufacturing processes.

LEADERSHIP EXPERIENCE

Muslim Student Association (MSA), *Member/Volunteer* — Aug 2023 – Present

- Coordinated 10+ community service and outreach events, collaborating with 20+ volunteers to increase member engagement and strengthen community involvement.

SKILLS

- **Software:** *SolidWorks, Creo, AutoCAD, Ansys Workbench, MATLAB, Excel, Python, Microsoft Office, Google Workspace*
- **Engineering:** SLA/FDM *3D Printing, CNC, Manual Machining, GD&T,* DFM, Drawings, BOM Creation